- ***Candidates should be able to*** :

  - *Describe and distinguish between **progressive longitudinal** and **transverse** waves.*

  - *Define and use the terms **displacement, amplitude, wavelength, period, phase difference, frequency and speed of a wave.***

  - ***Derive** from the definitions of speed, frequency and wavelength, the wave equation :*

    $$v = f\lambda$$

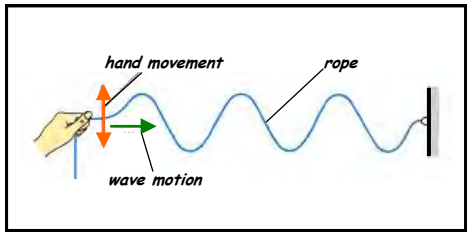
  - *Select and use the wave equation :* $v = f\lambda$

  - *Explain what is meant by **reflection, refraction** and **diffraction** of waves such as sound and light.*

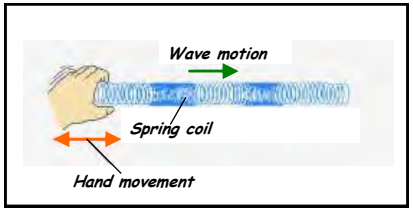
- ***PROGRESSIVE WAVES***

- *With the exception of electromagnetic waves, which do not need a material substance for their transmission, all other wave motions (e.g. sound waves, water waves etc.) are disturbances which pass through a substance.*

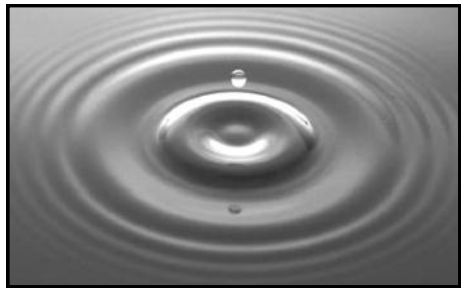
- *A series of waves (similar to those produced on a water surface) can be generated in a long piece of rope which is fixed at one end and moved repeatedly at right angles to its length (as shown in the diagram opposite).*



- *Waves (similar to sound waves) are produced in a spring coil which is fixed at one end while the other end is repeatedly moved as shown in the diagram opposite.*



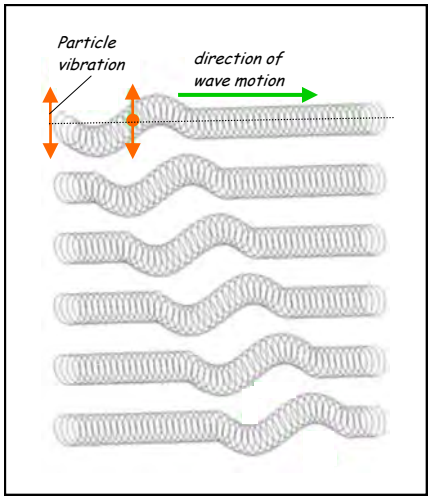
- *A pebble dropped onto a still water surface causes a vibration which produces concentric, circular waves which spread outwards across the water surface.*



- *In each of the wave examples considered, the wave disturbance is initiated at a particular point and this sets the particles of the substance at that point into vibration. This causes neighbouring particles to vibrate in the same way and so the wave progresses through the substance.*

In the case of a single wave pulse initiated by an up and down movement of the end of a springy coil, the motion of the first ring causes the next ring to move and so on.

In this way, the wave travels from one end of the coil to the other as shown in the diagram opposite.



- *All the waves considered so far are :*

PROGRESSIVE WAVES

Waves which travel through a material or through a vacuum and transfer energy from one point to another.

- ***CLASSIFICATION OF WAVES*** *2*

- *Waves can be :*

MECHANICAL — OR — **ELECTROMAGNETIC**

Waves which need a substance for their transmission.

(e.g. sound waves can be transmitted Through air, water and steel, but not Through a vacuum).

EXAMPLES

- *Water waves*
- *Sound waves*
- *Waves along a spring coil or rope*
- *Seismic waves*

*Waves which **DO NOT** need a substance for their transmission*

(e.g. light waves travel from stars and galaxies travel through empty space to reach us here on Earth).

EXAMPLES

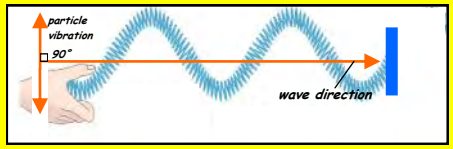
- *Gamma-rays*
- *X-rays*
- *Ultra-violet (UV)*
- *Visible light*
- *Infra-red (IR)*
- *Microwaves*
- *Radio waves*

Wave Motion

- *Waves can be :*

TRANSVERSE — OR — **LONGITUDINAL**

*Waves in which the particles of the medium in which a wave is moving vibrate **perpendicular** To the direction of wave travel.*

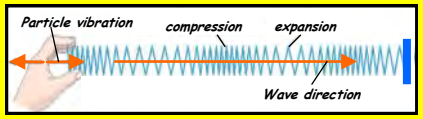


*A spring coil can be used as shown above to observe **TRANSVERSE** wave motion. The coil is fixed at one end and the free end is continuously vibrated in a direction **PERPENDICULAR** to the axis of the coil.*

TRANSVERSE WAVE EXAMPLES

- ***Water waves***
- ***Secondary seismic waves***
- ***All electromagnetic waves***

*Waves in which the particles of the medium in which a wave is moving vibrate **parallel** to the direction of wave travel.*



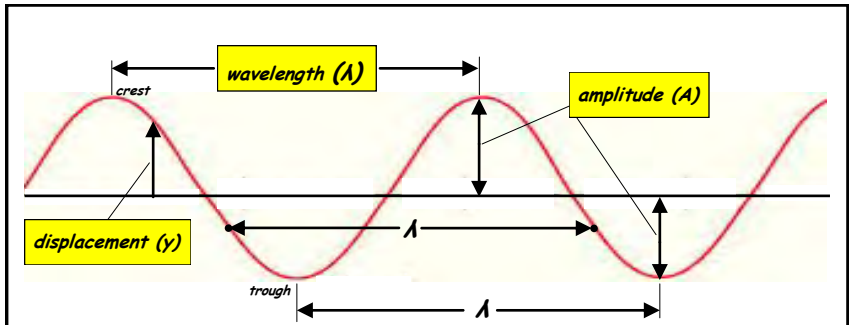
*A spring coil can be used as shown above to observe **LONGITUDINAL** wave motion. The coil is fixed at one end and the free end is continuously vibrated in a direction **PARALLEL** to the axis of the coil.*

LONGITUDINAL WAVE EXAMPLES

- ***Sound waves***
- ***Primary seismic waves***

- *TERMS ASSOCIATED WITH WAVES*

3



DISPLACEMENT (y) / metre (m)

*The **distance** and **direction** of a vibrating particle in a wave from its undisturbed position.*

AMPLITUDE (A) / metre (m)

*The **maximum displacement** of any particle in a wave from its undisturbed position.*

WAVELENGTH (λ) / metre (m)

*The distance between two **consecutive points** on a wave which are **in phase** with each other (i.e. the vibrating particles at these points have the same displacement and velocity).*
*It is the distance between two consecutive **crests** or **troughs**.*

PERIOD (T) / second (s)

The time taken for one complete wave to pass a fixed point.

OR

The time taken for one complete oscillation of a particle in the wave.

FREQUENCY (f) / hertz (Hz)

The number of complete waves passing a fixed point per second.

OR

The number of complete oscillations per second of a particle in the wave.

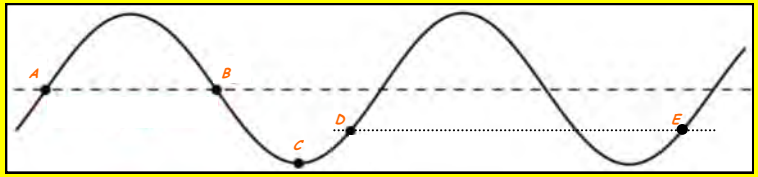
1 Hz = 1 vibration per second.
1kHz = 10^3 Hz **1 MHz** = 10^6 Hz.

PHASE DIFFERENCE (Ɇ) / degrees or radians

The phase difference between two vibrating particles in a wave is the fraction of a cycle between the vibrations of the two particles.

*Phase difference is measured in **DEGREES** or **RADIANS**.*

1 cycle = 360° = 2π radians



- *In the diagram above, particles at points **D** and **E** which are one wavelength apart, vibrate **in phase** with each other. The **phase difference** between the partcles at these two points is **360° (=2π rads)** (which is the same as **0°**).*
- *Particles at points **A** and **B** which are $\frac{1}{2}$ a wavelength apart, vibrate in **antiphase**. The **phase difference** between the particles at these two points is **180° (π rads)**.*
-

C

Wave Motion

WAVE SPEED (v or c) / metre per sec (m s^{-1})

This is the speed with which energy is transmitted by a wave.

WAVE SPEED EQUATION

*Consider a wave having **wavelength (λ), period (T), frequency (f)** and moving with **speed (v)***

In time (T) the wave will travel a distance (λ).

wave speed, v = $\frac{\text{distance travelled}}{\text{time taken}} = \frac{\lambda}{T}$

And since, T = 1/f, $v = \frac{\lambda}{1/f}$

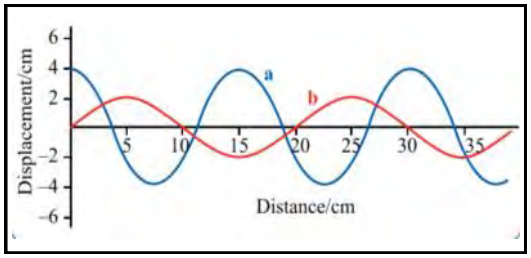
Therefore :

$$v = f\lambda$$

(m s^{-1}) (Hz) (m)

- **PRACTICE QUESTIONS**

1 Determine the **amplitude** and **wavelength** of waves **a** and **b** shown in the diagram opposite.



2 (a) Sound waves travel at a speed of **340 m s^{-1}** in air at **20°C**. If the frequency of these waves **is 2.5 kHz**, what is their **wavelength ?** *5*

(b) Infra-red electromagnetic waves travel at a speed of **3.0×10^8 m s^{-1}**. Calculate the **frequency** of these waves if their wavelength is **1.2×10^{-6} m**.

3 A wave of frequency **2.5 Hz** travels along a stretched cable at a speed of **20 m s^{-1}**. Calculate the **phase difference** between two points on the cable which are **2.0 m** apart.

Express your answers in : (a) **Degrees**, (b**) Radians** and (c) **Fractions of a wavelength.**

4 Calculate the **speed** of a progressive wave of frequency **350 Hz**, if the least distance between two points on the wave having a phase difference of **π/6 rad** is **0.08 m.**

5 A laser emits a single burst of light which lasts for a time of **0.02 μs.** If the wavelength of the light is **500 nm,** calculate the **number of complete waves emitted** by the laser.

6 Using graph paper, draw two waves, **X** and **Y**, such that **Y** has **twice the wavelength** and **a quarter of the amplitude** of **X.**

7 Use the data given below to calculate the **frequency range of light** detectable by the human eye.

Speed of light in air = **3.0×10^8 m s^{-1}.**

Approximate range of visible light wavelengths is **400 nm (violet)** to **700 nm (red).**